The Boolean Satisfiability (SAT)

- SAT is a problem of deciding (requires a yes/no answer) if there is an assignment to variables of a Boolean formula such that the formula is satisfied.
- That is, whether there is some combination of the (binary) values of the variables that will give 1.
- A given Boolean expression, determining if there exists a truth assignment to its variables for which the expression is *true* is defined as the satisfiability problem.

  ## EXAMPLE

  - Consider the formula $(a \vee b) \wedge (\neg a \vee b)$
  - Can we assign values such that the expression is TRUE?
  - Seems simple enough, but no known deterministic polynomial time algorithm exists
  - Easy to verify in polynomial time

CNF (Conjunctive Normal Form)

- When we're discussing the SAT problem, it's essential to know about the conjunctive normal form (CNF).
- A Boolean expression is said to be in CNF form if it's a conjunction of a set of clauses, where each clause is defined as a disjunction (logical OR) of literals. We can define a literal as a variable or a negation of a variable.

  CNF:$C1 \wedge C2 \wedge C3 \wedge \ldots$ .CK

Ci:x1∨x2∨… xn

3-SAT

- The Boolean satisfiability problem where just three clauses are involved.
- In the 3-SAT problem, each clause has at most 3 literals.
- 3-SAT asks whether it is possible to solve the Boolean satisfiability problem provided that there are at most 3 variables between each pair of parentheses in the Boolean formula.

E.g. (x1 ∨ x2 ∨ x3) ∧(x4 ∨ x5 ∨ x6)

SAT Problem :- Given a set of clauses C1, C2, . . . , Cm in CNF form, where Ci contains literals from x1, x2, . . . , xn, problem is to check if all clauses are simultaneously satisfiable.

3SAT problem: - Given a set of clauses C1, C2, . . . , Cm in 3CNF from variables x1, x2, . . . , xn problem is to check if all the clauses are simultaneously satisfiable.

3-SAT := {Satisfiable 3CNF formulae }